

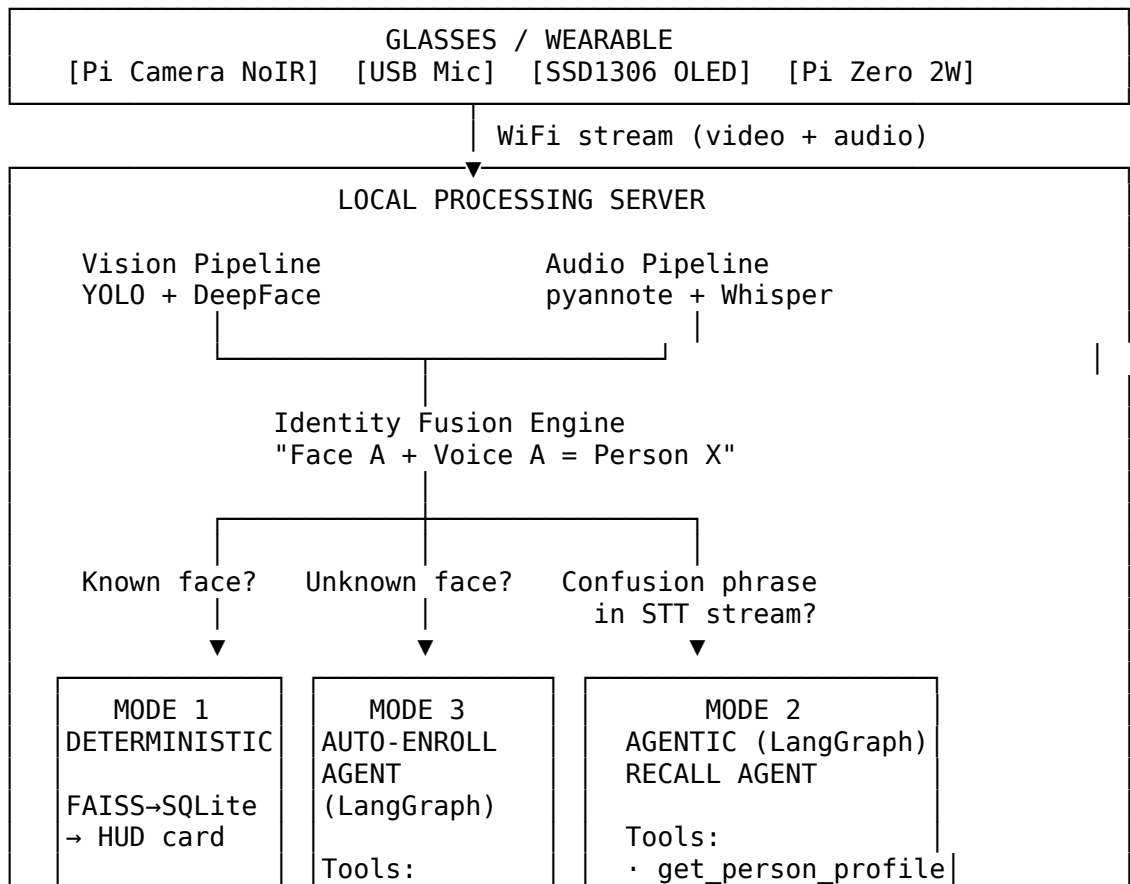
memorylens

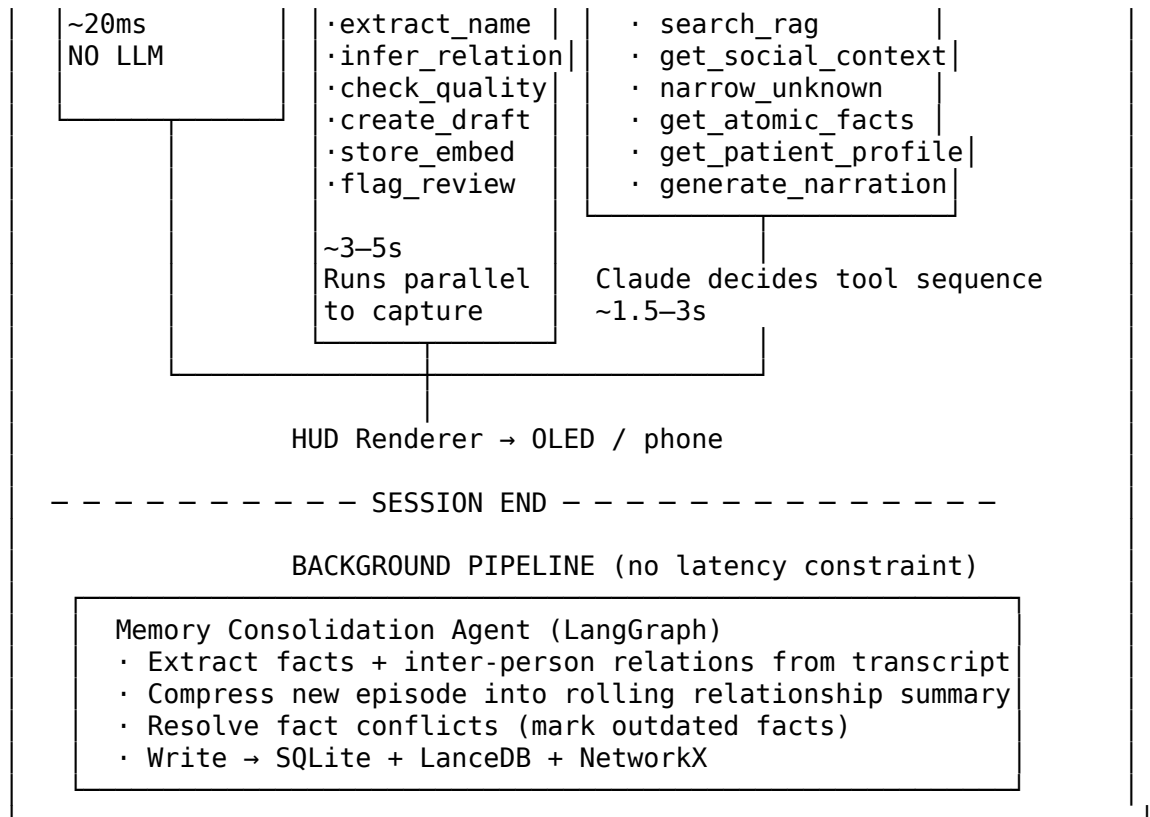
Memory assistance system for dementia patients using multimodal identity recognition and conversational AI.

Smart Glasses Identity & Recall Architecture

A wearable AI system built on Raspberry Pi, routing audio-visual streams to a local processing server for real-time identity resolution, agentic recall, and post-session memory consolidation.

System Architecture





□ Component Breakdown

1. Hardware Layer (Wearable)

The edge device captures real-time environment data and displays context back to the user.

- **Microcontroller:** Pi Zero 2W
- **Vision:** Pi Camera NoIR
- **Audio:** USB Microphone
- **Display:** SSD1306 OLED

2. Local Processing Server (Real-Time)

Data is streamed via WiFi to a local server for heavy lifting.

- **Vision Pipeline:** Utilizes **YOLO** (object/face detection) + **Deep-Face** (facial recognition/embedding).

- **Audio Pipeline:** Utilizes **pyannote** (speaker diarization) + **Whisper** (Speech-to-Text).
- **Identity Fusion Engine:** Merges sensory inputs to create a unified identity profile.

✂ Operational Modes

Feature	Mode 1: Deterministic	Mode 2: Agentic Recall	Mode 3: Auto-Enroll
Trigger	Known face recognized	Confusion phrase in STT stream	Unknown face detected
Architecture	AISS → SQLite → HUD	LangGraph + Claude	LangGraph
Latency	~ 20ms (Ultra-fast)	~1.5–3.0s	~3–5s (Runs parallel)
LLM Usage	None	Yes (Decides tool sequence)	Yes (Agentic drafting)
Output	Instant HUD Identity Card	Contextual memory / facts on HUD	Background profile creation

▢ Background Pipeline (Post-Session)

When the active session ends, the **Memory Consolidation Agent** organizes and permanently stores the session data.

1. **Extraction:** Pulls new facts and inter-person relations from the transcripts.
2. **Compression:** Summarizes the new episode into a rolling relationship summary.
3. **Conflict Resolution:** Identifies conflicting information and marks outdated facts.
4. **Database Writes:** Pushes structured knowledge to SQLite, LanceDB, and NetworkX.